

Session Objectives

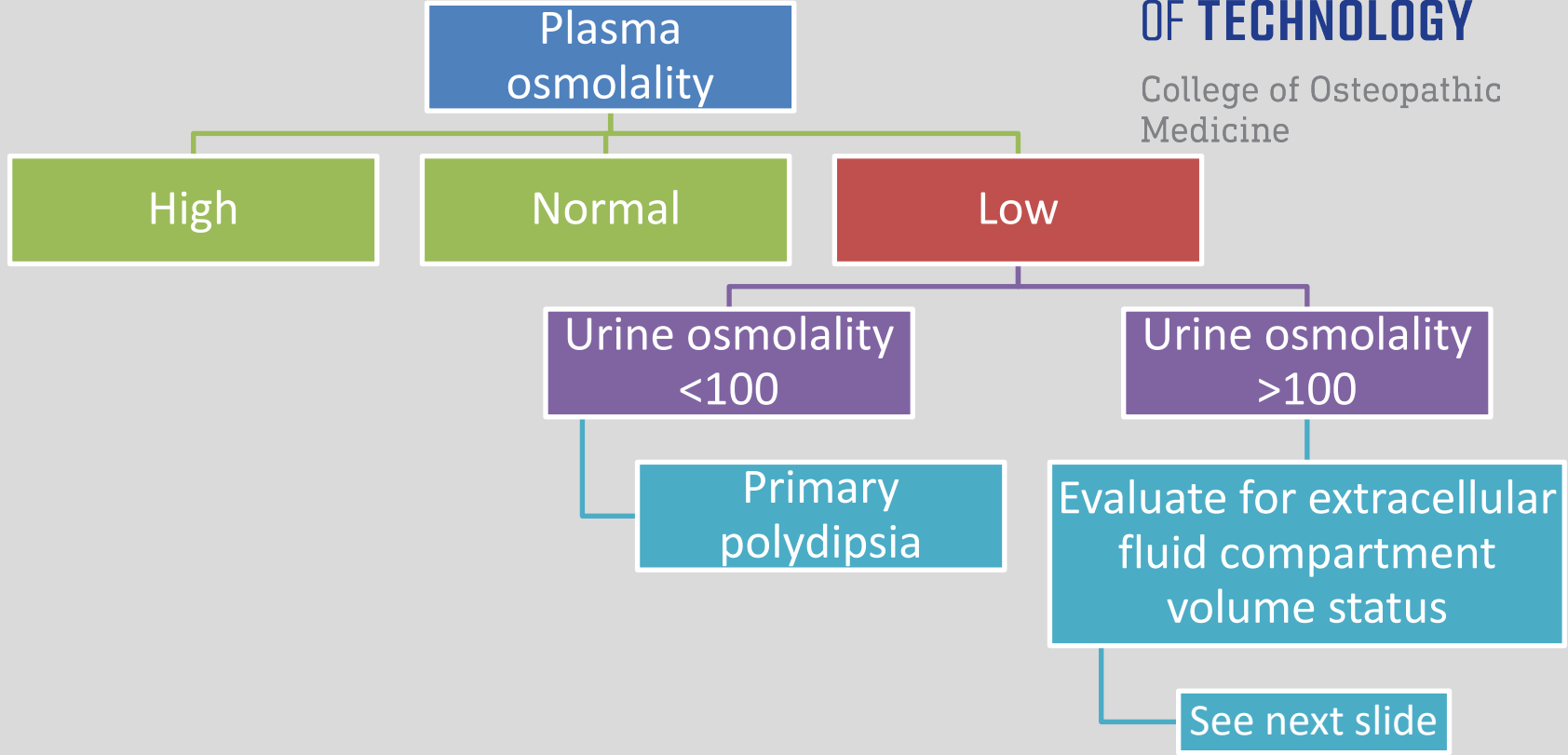
- Distinguish the clinical features and complications of hyponatremia.
- Illustrate the pathophysiology of hyponatremia.
- Judge the causes of hyponatremia.
- Explain the treatment of hyponatremia.
- Explain the syndrome of inappropriate antidiuretic hormone secretion.

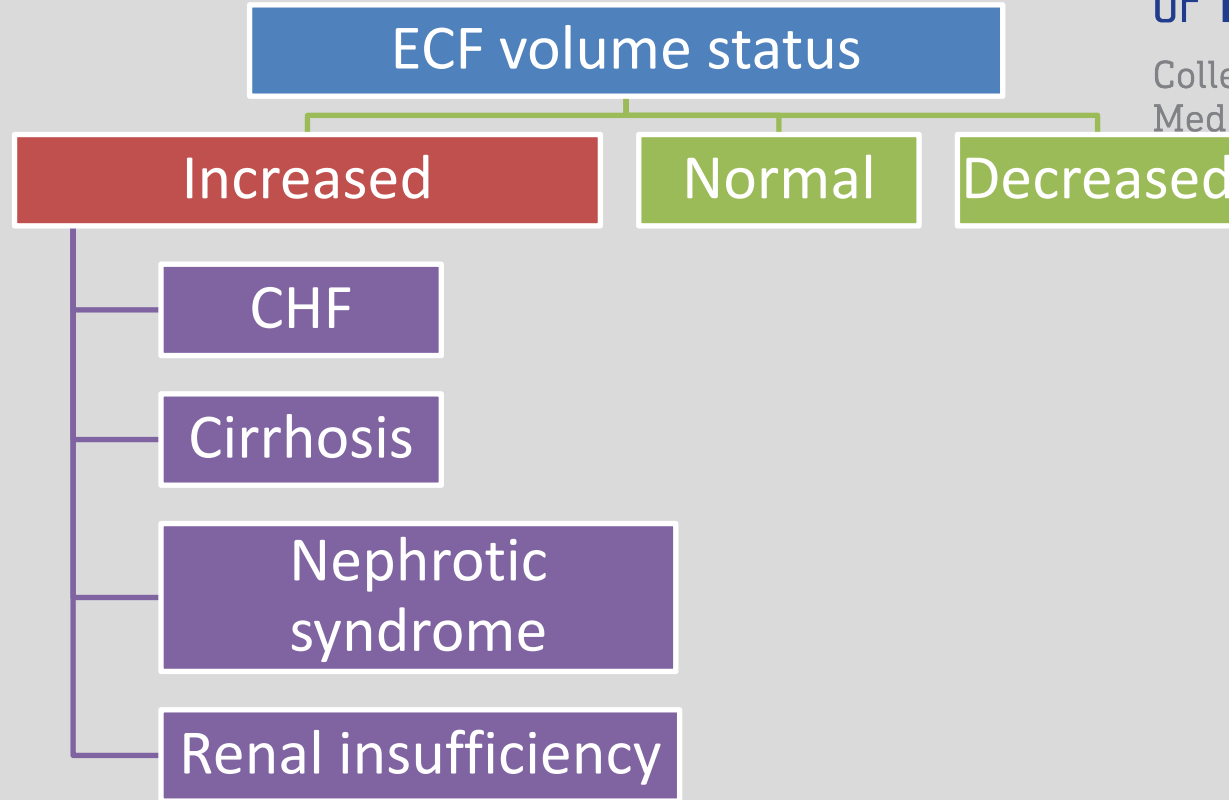


Hyponatremia Classification

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Medicine



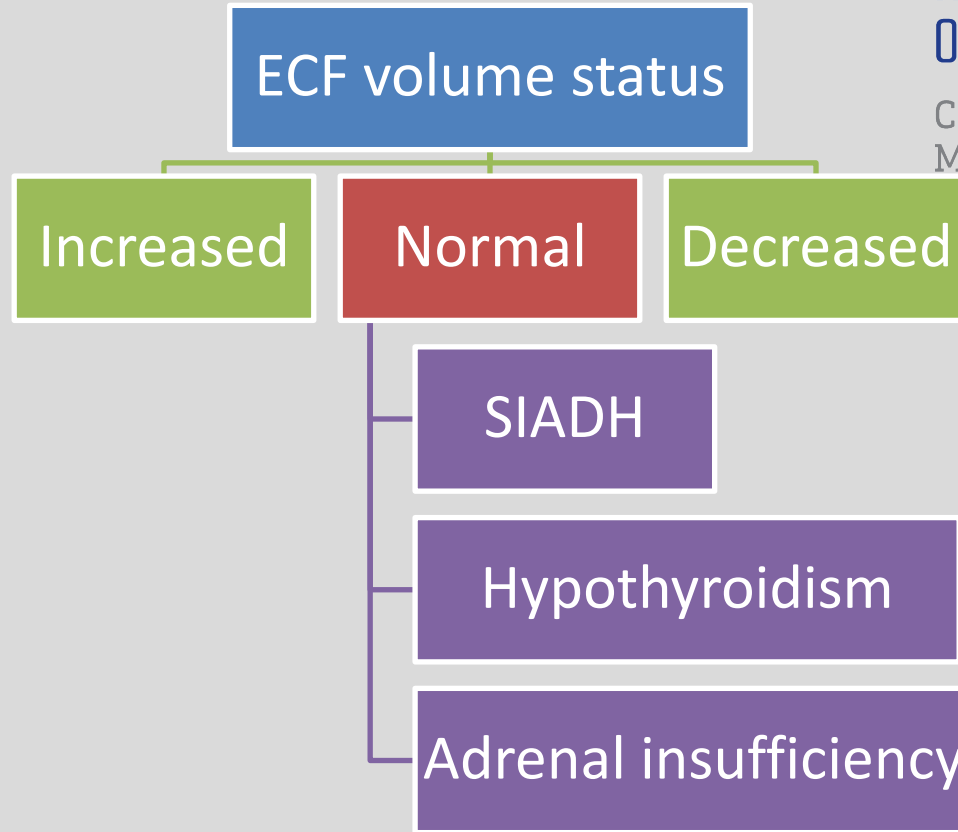




Hypotonic Hyponatremia with: Low Plasma osmolality

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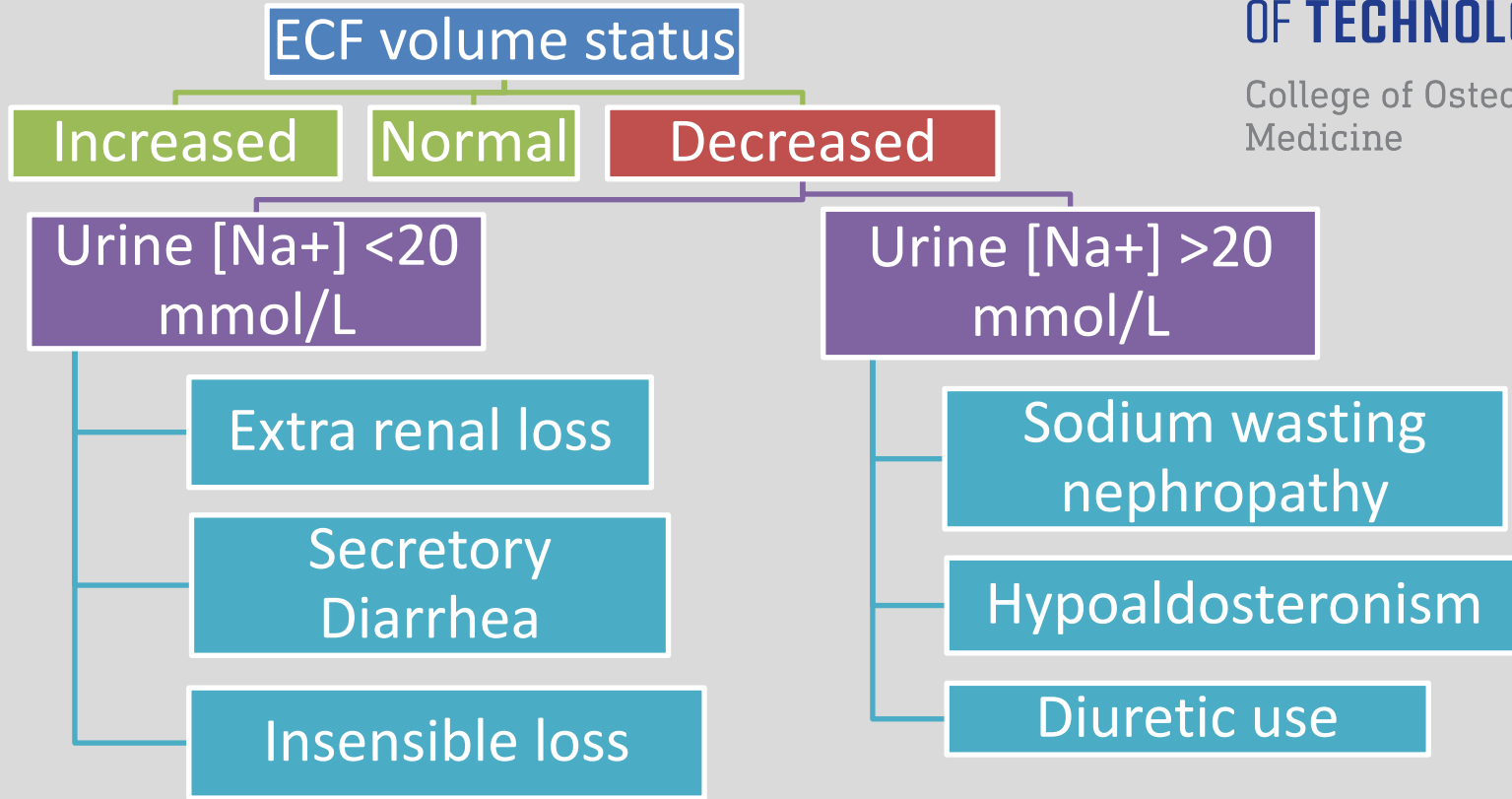




Hypotonic Hyponatremia with: Low Plasma osmolality

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Question

A 49-year-old female presents to emergency department with complaints of fatigue, a headache and nausea. Laboratory studies reveal the following: sodium is 129 mEq/L (normal 135-145 mEq/L), serum osmolality is 250 mOsm/kg (normal 275 to 290 mosmol/kg). What test should be next in our workup to differentiate the causes of hyponatremia?

- A. ADH levels
- B. Urine osmolality
- C. Stool culture
- D. Echocardiogram



Question

A 49-year-old female presents to emergency department with complaints of fatigue, a headache and nausea. Physical examination reveals no edema and good skin turgor. Laboratory studies reveal the following: sodium is 129 mEq/L (normal 135-145 mEq/L), serum osmolality is 250 mOsm/kg (normal 275 to 290 mosmol/kg), urine osmolality is 50 mOsm/kg (normal 500-800 mOsm/kg).

Which of the following is the most likely cause of this patient's hyponatremia?

- A. Hyperlipidemia
- B. Primary polydipsia
- C. Hyperglycemia
- D. Cirrhosis
- E. Sodium wasting nephropathy



Question

Which of the following is the most likely pathophysiology behind this patient's hyponatremia?

- A. Increased plasma volume of a substance not contributing to plasma osmolality.
- B. Increased thirst causing water intoxication which overwhelms the body's capacity to excrete water.
- C. Increased plasma osmolality induced by a substance drawing water out of the cells.
- D. A decrease in oncotic pressure resulting in markedly increased vasodilatation and decreased arterial blood volume.
- E. A primary defect in sodium chloride reabsorption in the medullary thick ascending limb of the loop of Henle.

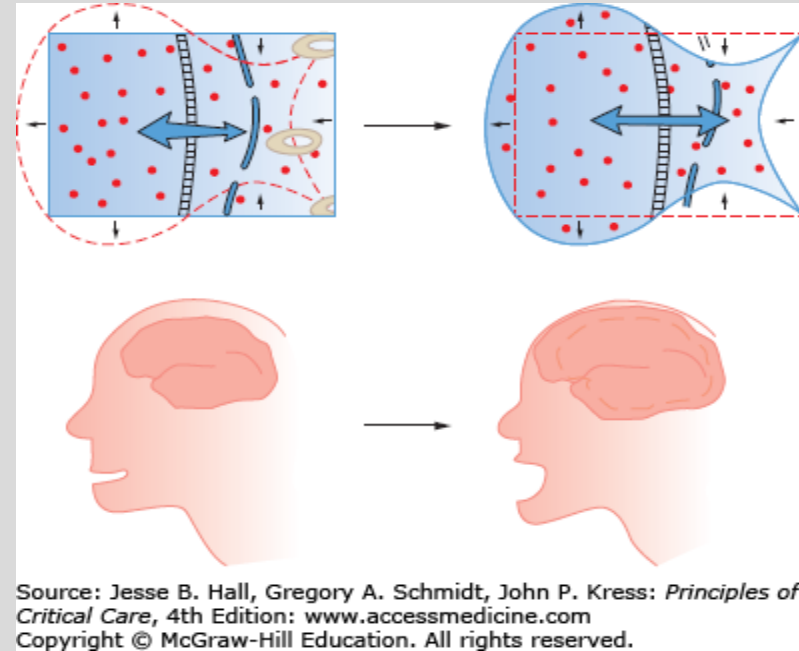
Cerebral Edema

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Acute hyponatremia

- Extracellular fluid hypotonic (lot of water lower solute) compared to intracellular fluid of neurons (less water higher solute).
- Water drawn in by osmosis
- **Cerebral edema via herniation**



Overly aggressive therapy

- Hypertonic saline after adaptation occurs.
- Extracellular fluid more concentrated than intracellular fluid.
- Draws water out of brain cells in **osmotic demyelination**.



Treatment Hyponatremia

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- Duration of hyponatremia
 - Acute - first 24 hrs
 - Chronic – more than 48hrs
- Severity of symptoms
 - Or asymptomatic
- Degree of hyponatremia
 - Mild 130-135 meq/L serum sodium
 - Moderate 121-129 meq/L serum sodium
 - Severe 120 meq/L or less.

Setting of treatment

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- Hospitalize – pts with severe hyponatremia and symptomatic moderate hyponatremia
- Outpatient – mild hyponatremia and asymptomatic moderate hyponatremia

Treatment

- Emergent therapy – Aggressive therapy to increase sodium levels if:
 - Severe symptoms
 - Acute symptomatic hyponatremia
 - Self induced water intoxication



Question

An 88-year-old male presents to the emergency department with a complaint of bilateral lower extremity edema. He has a longstanding history of cirrhosis and hyponatremia. The resident notices his sodium level is 119 mEq/L (normal 135 and 145 mEq/L) and administers copious hypertonic saline. Two days later the patient is comatose and eventually dies. On autopsy which of the following brain structures is most likely to be affected?

- A. Pons
- B. Cerebellum
- C. Medulla
- D. Pre-frontal cortex
- E. Parietal lobe

Case scenario

- 78-year-old man was admitted with pneumonia and hyponatremia.
- Plasma Na^+ concentration was initially 129 meq/L, decreasing within 3 days to 118–120 meq/L despite fluid restriction to 1 L/d. A chest CT revealed a right 2.8×1.6 cm infrahilar mass and postobstructive pneumonia.
- The patient was an active smoker. Past medical history was notable for laryngeal carcinoma treated 15 years earlier with radiation therapy, renal cell carcinoma, peripheral vascular disease, and hypothyroidism.

Case scenario

- On review of systems, he denied headache, nausea, and vomiting. He had chronic hip pain that was managed with acetaminophen with codeine.
- Other medications included cilostazol, amoxicillin/clavulanate, digoxin, diltiazem, and thyroxine.
- He was euvolemic on examination, with no lymphadenopathy and a normal chest examination.

SIADH

- Impaired water excretion
- Inability to suppress ADH
- ADH causes concentrated urine increase total body water.
- Low plasma sodium concentration

Key Points

- Key points in management of the patient with hyponatremia
 - Diagnose the cause
 - Clinical history / findings / lab test
 - Determine a treatment plan
 - How to treat
 - How fast to treat
 - What to monitor
 - Assess need for follow up
 - When do you expect improvement
 - When do you need to next evaluate the patient
 - Anticipate complications from treatment



Question

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A 55-year-old female presents to ED with complaints of fatigue, a headache and nausea. PE reveals no edema and good skin turgor. Lab studies reveal sodium is low, serum and urine osmolality are normal. What is the diagnosis?

- A. Hyperlipidemia
- B. Hyperglycemia
- C. Cirrhosis
- D. Hypothyroidism